

25PY101 (S2): Engineering Physics

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Mock pre M1T1 (Set II)

Instructions:

1. Answer all questions.
 2. Total marks – 10 M.
 3. Charge of electron is 1.602×10^{-19} C.
 4. Mass of electron is 9.1×10^{-31} kg.
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1 Metals and Semiconductors – Hall effect

- (a) A copper strip 2 cm wide and 1 mm thick is placed in a magnetic field with $B = 1.5$ T. If a current of 200 A is set up in the strip, calculate the Hall voltage that appears across the strip. Assume $R_H = 6 \times 10^{-7} \text{ m}^3 \text{ C}^{-1}$. [1 M]
- (b) What is the Hall coefficient of an intrinsic semiconductor? [1 M]
- (c) Hall coefficient of copper is smaller than that of silicon with same dimensions. Explain. [1 M]
- (d) An electric field of 100 V m^{-1} is applied to a sample of n-type semiconductor whose Hall coefficient is $-0.0125 \text{ m}^3 \text{ C}^{-1}$. Determine the current density in the sample, assuming $\mu_e = 0.6 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$. [2 M]

2 Quantum mechanics – de Broglie's hypothesis

- (a) State the de Broglie's hypothesis. [1 M]
- (b) Find the expression for the de Broglie wavelength of electron in each of the below three cases.
 - (i) The kinetic energy of an electron is K . [1 M]
 - (ii) The temperature of an electron is T . [1 M]
 - (iii) The electron under rest is accelerated under a potential difference of V . [1 M]
- (c) An electron and a proton have the same kinetic energy. Which of them has a greater de Broglie wavelength? [1 M]

End of Test